Kangaroo: Demand-Driven Cross-Environment Navigation for Macro-Micro Vehicles Enhanced by Large Language Models

Abstract

Demand-driven navigation is a crucial research area in mobile robotics, with broad service applications across diverse real-world scenarios such as logistics, inspection, and agricultural automation. Although some heterogeneous multi-agent systems (HMAS) have demonstrated navigation capabilities in various isolated environments, existing research has not explored robot navigation across continuous multi-scale scenarios. Moreover, they cannot achieve both scene and demand adaptability. To this end, we propose *Kangaroo*, a demand-driven cross-environment HMAS navigation framework designed for Macro-Micro Vehicles (MMVehicles). First, we introduce a novel motion primitive generation strategy, the Iterative Feedback Planner, which is assisted by large language models and guided by human interaction, rule-based feedback, and prior knowledge. This method enhances demand instruction comprehension and significantly boosts coordination among multiple agents. In addition, we design a navigation controller for MMVehicles equipped with a multi-sensor fusion module for Simultaneous Localization and Mapping (SLAM) and collaborative navigation. These modules empower MMVehicles with real-time environment perception and action generation, facilitating seamless transitions across diverse environments from expansive open fields to tightly confined spaces. We conduct experiments in multi-scale real-world scenarios. The results show that *Kangaroo* achieves an over 85% success rate in VastHall-TinyRoom and 78% in OutdoorRoad-NarrowPath scenarios when processing demands across four complexity levels. A demonstration video is available at https://youtu.be/KHXHtGMmNqc.

Keywords:
Demand-Driven Navigation, Cross-Environment, Heterogeneous Multi-Agent System, Large Language Model, Hybrid Simultaneous Localization and Mapping

1. Introduction

Navigation [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12] for mobile robots guided by natural language demands represents a pivotal research topic in Embodied AI. Robots are expected to interpret abstract human demand instructions (e.g., "I'm thirsty") and execute real-time actions to fulfill these needs within a given environment, under the limited computational resources of edge devices. Recent advancements in large language models (LLMs), simultaneous localization and mapping (SLAM), and multi-agent systems (MAS) have significantly enhanced the capabilities of Demand-Driven Navigation, enabling its deployment in service applications such as household service robots[13, 1], Harsh Environment Rescue Robot[14, 15], and outdoor logistics robots[16, 17]. However, existing demand-driven navigation methods are confined to isolated environments, with limited focus on cross-environment adaptation, particularly seamless transitions between varying-scale scenes during continuous movement. This calls for a unified approach that simultaneously addresses both demand-driven adaptability and cross-environment generalization with an effective collaborative mechanism of edge devices.

Demand-Driven Cross-Environment Navigation presents an inevitable challenge for real-world robot systems, as different navigation scenarios introduce distinct requirements and environmental characteristics. While intrinsic variations (e.g., color, texture) across scenarios can be mitigated through dataset augmentation or adaptive feature extraction algorithms, continuous scale variations and cross-environment transitions (e.g., from outdoor roadways to narrow passages) present more profound challenges. These transitions often impose conflicting kinematic demands, as expansive areas prioritize high-speed mobility while confined spaces require precise maneuverability, which cannot be effectively addressed by a single agent or homogeneous agents. The fixed hardware morphology of current systems further restricts adaptability to scale-variant environments, motivating our proposal of a heterogeneous multi-agent system (HMAS)[18] called Macro-Micro Vehicles (MMvehicles) for Demand-Driven Cross-Environment Navigation. To achieve this, our system must satisfy two core requirements: (1) understanding demand-driven instruction for intelligent high-level task allocation, and (2) enabling precise collaborative navigation of the HMAS during cross-environment transitions.

Vision-Language Models (VLMs) and Large Language Models (LLMs)



Figure 1: **MMVehicles in Demand-Driven Cross-Environment Scenarios.** The MMVehicles system receives demand instructions and environmental knowledge. Under predefined rule constraints, a large language model (LLM) performs global motion primitive planning. SLAM is employed for perception and localization, while a motion controller executes the planned trajectories and manages the control of MMVehicles. These components enable the system to achieve seamless multi-scale transitions across environments like outdoor-indoor scenarios.

enable robots to interpret environments and language instructions for action generation and task scheduling. Retrieval-Augmented Generation (RAG)[19] enhances this by integrating external knowledge retrieval, improving decision-making accuracy and adaptability. [1, 20] leverage VLMs to coordinate navigation among multi-agents. However, their system involves only homogeneous agents and lacks any verification mechanism for VLN, making it difficult to ensure reliability. These limitations highlight the necessity for an LLM-based framework that can support reliable coordination within HMAS while enabling verified natural language understanding. Besides, end-to-end vision-language navigation (VLN) methods often struggle with accuracy due to limited data and environmental variability. In contrast, SLAM-based approaches exploit physical constraints and prebuild maps for more reliable spatial understanding. 2D SLAM methods (e.g., Cartographer [21], GMapping [22]) work well indoors but are limited by planar assumptions. 3D SLAM (e.g., LVI-SAM [23], HDL-Graph-SLAM [24]) achieves high navigation accu-

racy but suffers from texture perception limitations and high hardware costs. Moreover, the adaptability of these methods to different scales of environments depends on the sensors and optimization methods. As a result, hybrid or complementary approaches are necessary to address the challenges posed by multi-scale and heterogeneous environments.

To our knowledge, no prior work has addressed the challenge mentioned above. We ask: **Can LLM-assisted HMAS effectively resolve navigation challenges in cross-environment, demand-driven scenarios?**

To this end, we propose *Kangaroo*, a demand-driven cross-environment Macro-Micro Vehicles Navigation system. Its hardware platform consists of a high-mobility MacroVehicle and a highly flexible MicroVehicle, inspired by the Kangaroo and its joey. For demand understanding, we introduce an external knowledge base to abstract environmental content and integrate multi-dimensional rule constraints (e.g., primitive consistency, temporal logic, hardware specifications, and edge case validation) with the human-AI iterative interactions of large language models (LLMs). This approach aims to address the challenges posed by abstract demands and the inherent instability of LLMs. For precise cross-environment navigation, we employ a hybrid SLAM-based navigation method with multi-sensor fusion on resource-constrained edge devices, ensuring real-time perception and planning across diverse scene features. Furthermore, a four-level collaboration mechanism incorporating a dual-channel hybrid network, cooperative perception, visual-Lidar docking, and LLMs high-level planning ensures efficient cooperation and information sharing between agents in cross-environments.

Our main contributions are as follows:

- We propose *Kangaroo*, the first HMAS for demand-driven cross-environment navigation, integrating hybrid SLAM-based navigation with macro-micro vehicle coordination mechanisms, guided by LLMs.

- We propose the Iterative Feedback Planner (IF-Planner), a LLM-based motion primitive generator. Enhanced by an external knowledge base, it employs an Interaction Module for intent parsing and a Rule Validator for constraint compliance, enabling precise natural language instruction interpretation and dynamic HMAS coordination control.

- We develop a Macro-Micro Vehicles hardware platform and conduct real-world experiments. *Kangaroo* achieves an 85% success rate in

VastHall-TinyRoom and 78% in OutdoorRoad-NarrowPath when handling instructions at four demand levels.

We demonstrate that the proposed LLM-assisted MMVehicles system enhances the adaptability and efficiency of navigation in cross-environments, offering a scalable solution for human-AI collaborative multi-agent service applications.

2. Related Work

2.1. Multi-Agent System

A multi-agent system (MAS) [18, 25, 26, 27, 28, 29]refers to a system composed of multiple intelligent agents, which can be either homogeneous or heterogeneous, that interact with each other to achieve a common objective through collaboration.

Ye et al. [20] propose a neural bipartite graph matching method that achieves high efficiency in homogeneous robotic systems. In parallel, Shen et al. [7] introduce a multi-agent path planning approach powered by large language models (LLMs), enabling agents to reason over structured instructions for navigation tasks. While both methods exhibit promising performance within structured environments, they are inherently limited to predefined scenarios and lack adaptability across heterogeneous or dynamic domains. To support distributed decision-making in multi-agent systems, several works have focused on communication mechanisms across multi-edge devices. For instance, [30, 31, 32] design effective inter-device communication protocols, while [33, 34] propose efficient schemes for task allocation and resource scheduling. These methods collectively establish the communication infrastructure necessary for collaborative interactions in distributed systems. Beyond communication, Giribet et al. [28] implement aerial-ground collaboration between unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) through non-inertial model predictive control (MPC). However, these methods are unable to comprehend human demand semantics, resulting in a disconnect between high-level semantic interpretation and low-level control decision-making.

There is an urgent need to design a heterogeneous multi-agent system to address the dual challenges of dynamic scene adaptation and coordination in cross-environment navigation.

2.2. SLAM

Simultaneous Localization and Mapping (SLAM)[21, 35, 36] is a crucial algorithm for autonomous navigation in mobile robots, enabling them to estimate their poses while incrementally constructing environment maps in unknown settings. SLAM has evolved from 2D mapping to 3D representations and transitioned from single-sensor frameworks to multi-sensor fusion approaches.

Traditional 2D SLAM methods, such as GMapping[22] and Cartographer[21], rely on 2D LiDAR and employ particle filtering or graph optimization to achieve efficient localization and occupancy grid mapping. These methods are predominantly suited for structured indoor environments but are limited to planar perception, rendering them ineffective in unstructured or 3D-rich scenarios. To address this, vision-based SLAM approaches like ORB-SLAM3[36] leverage textural features and visual-inertial fusion for 3D mapping. However, their performance degrades under illumination variations, restricting applicability to well-lit environments. Meanwhile, 3D LiDAR-based SLAM (e.g., HDL-Graph-SLAM[24], LIGO[35], Fast-LIVO2[37]) demonstrates strong performance in large-scale outdoor environments due to its robust 3D perception, but its high hardware costs make it impractical for small-scale settings.

Although SLAM algorithms provide highly accurate localization for navigation tasks, their performance often varies significantly across different environments. Moreover, the lack of effective cross-environment coordination mechanisms limits their scalability and robustness in complex, dynamic scenarios. These challenges highlight the need for further research and improvement.

2.3. LLM-assisted Navigation

The advancement of embodied intelligence [38, 39, 40, 41, 42] has significantly contributed to bridging the gap between high-level human instructions and low-level robot control. Among its key applications, demand-driven navigation plays a critical role by enabling robots to understand and execute natural language commands, while incorporating commonsense reasoning to support more natural and robust autonomous navigation.

LLM-based navigation methods, such as NaviLLM [5] NavRAG [19] and NavGPT2 [6], incorporate advanced semantic reasoning, external knowledge bases or task-planning mechanisms to interpret complex instructions and

adapt to dynamic environments. By modeling high-level linguistic commands into structured navigation strategies, these methods enable robots to operate more intelligently in unstructured settings. FastNav [3] employs LLMs for high-level task planning but relies on single-turn inquiries, limiting comprehension accuracy and planning reliability due to the lack of interactive verification. Additionally, environmental noise, sensor uncertainty, and sim-to-real discrepancies hinder system robustness and deployment.

Our approach enhances language understanding and scheduling accuracy through iterative feedback with human interaction and rule constraints. By integrating LLM-based methods with SLAM, we achieve high-precision, robust navigation.

2.4. Comparison of Navigation Strategies

We compare the performance of different approaches in demand-driven cross-environment navigation, assuming that SLAM algorithms inherently incorporate navigation modules. As shown in Table 2.4, both 2D and 3D SLAM are constrained by the fixed design of single-robot systems, making them less adaptable to multi-scale environments. Moreover, these methods rely on predefined navigation logic and are incapable of interpreting complex natural language instructions. Consequently, although SLAM-based approaches maintain high precision within their respective domains, they lack the flexibility required to handle diverse environments and dynamic tasks. In contrast, end-to-end navigation methods based on LLMs exhibit strong task comprehension capabilities. However, their accuracy is often limited by insufficient training data, environmental dynamics, and sensor noise. To address these limitations, we propose *Kangaroo*, which integrates the high precision of SLAM with the collaborative scheduling and language understanding capabilities of LLMs. Furthermore, *Kangaroo* incorporates a hierarchical Macro–Micro Vehicle system to overcome mechanical constraints across scales, enabling efficient and adaptable decision-driven navigation in multi-scale environments.

3. Kangaroo: Demand-Driven Cross-Environment Navigation for MMVehicles via IF-Planner

3.1. Overview

Our goal is to develop a general HMAS navigation system that enables robots to move efficiently through multi-scale complex physical environments

Nav. Type	Nav. Strategy	No Large Datasets	3D Feat.	Cross-Env.	Demand Driven	Acc.
2D	Cartographer [21]	✓	✗	✗	✗	✓
3D	HDL-Graph-SLAM[24]	✓	✓	✗	✗	✓
	LVI-SAM[23]	✓	✓	✗	✗	✓
	Fast-LIVO2[37]	✓	✓	✗	✗	✓
LLM-based	NaviLLM [5]	✗	✓	✗	✓	✗
	SPOC[43]	✗	✓	✗	✓	✗
	FastNav[3]	✗	✓	✗	✓	✓
Ours	**Kangaroo**	✓	✓	✓	✓	✓

Table 1: Qualitative Comparison of Different Methods.

and diverse demand scenarios. We develop a Macro-Micro Vehicles hardware platform and propose *Kangaroo*, a Demand-Driven Cross-Environment Navigation System guided by LLMs. Our system adopts a Main-Replica architecture, where the MacroVehicle serves as the main unit (central control and large-scale execution), while multiple micro vehicles function as replica units (small-scale execution). As illustrated in Fig. 2, our system comprises two key components: (1) IF-Planner, (2) MMVehicles Navigation Controller, and (3) MMVehicles Executor. The IF-Planner, deployed on the MacroVehicle, interprets nature language, generates motion primitive sequences, and coordinates operations between MacroVehicle and microVehicles. The Robot Navigation Controller is implemented on both MacroVehicle and MicroVehicle, integrating SLAM-based navigation algorithms and managing edge-device control to enable multi-scale perception and motion. MMVehicles Executor: The physical layer that executes motion commands. It receives instructions from higher-level modules and triggers mechanical actions via the onboard controller. This architecture ensures adaptive task scheduling and efficient navigation across complex physical and task environments.

3.2. Formulation

We formalize the IF-Planner-guided cooperative navigation problem for the Macro-Micro Vehicles (MMVehicles) system as follows.

At each time step t, the MMVehicles Navigation Controller computes the control outputs for both the MacroVehicle and microVehicle based on the real-time perception data C^t, the set of task-oriented strategies T_i, the MacroVehicle map M_{macro}, and the micro vehicle maps M_{micro}. This process

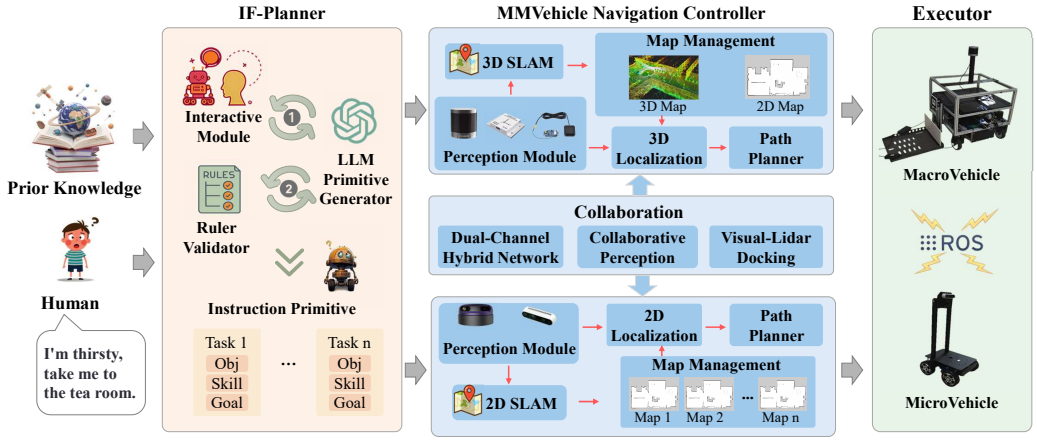


Figure 2: **Overview of the *Kangaroo* System Architecture.** The *Kangaroo* system is designed to address demand-driven navigation for heterogeneous robots in multi-scale environments. It takes human instructions and prior knowledge as inputs, with three core components: IF-Planner, MMVehicles Navigation Controller, and Execution Layer. The IF-Planner generates high-level motion primitives through intent understanding and rule-based reasoning to guide navigation. The Navigation Controller handles mapping, localization, and path planning, issuing low-level commands divided into MacroVehicle (3D SLAM) and MicroVehicle (2D SLAM) navigation. Collaboration among robots is achieved via a dual-channel hybrid network, cooperative perception, and visual-lidar docking. The Execution Layer uses ROS to execute commands, ensuring efficient navigation in complex environments.

is governed by the MMVehicles Navigation Controller policy π:

$$(U^t_{macro}, U^t_{micro}) = \pi(C^t, T_i, M_{macro}, M_{micro}) \tag{1}$$

where U^t_{macro} and U^t_{micro} denote the control commands for MacroVehicle and MicroVehicle, respectively. This formulation captures the language-conditioned multi-robot navigation problem under known environmental maps and dynamic perception inputs. The strategy set T_i is generated by IF-Planner, which interprets a natural language task description Q enriched with human feedback H. The IF-Planner utilizes prior knowledge Θ and conforms to a set of domain-specific constraints R to produce task-relevant strategies:

$$T = T_i \leftarrow \text{LLMs}(Q, H \mid \Theta, R) \tag{2}$$

while the MacroVehicle constructs global maps M_{macro} via MacroSLAM processing LiDAR/IMU/GPS data C_{macro}

$$M_{macro} \leftarrow \text{MacroSLAM}(C_{macro}) \tag{3}$$

Algorithm 1 Kangaroo

Input: Q, Θ, C, R
Output: U^m, U^c

$M_{macro} \leftarrow \text{MacroSLAM}(C)$
$M_{micro} \leftarrow \{M^1_{micro}, \ldots, M^n_{micro}\} \leftarrow \text{MicroSLAM}(C)$
while $\{Q,H\}$ is ambiguous **do**
  $H \leftarrow \text{Update}(H)$
end while
$T \leftarrow \{T_1, \ldots, T_n\} \leftarrow \text{IF-Planner}(Q, H|\Theta, R)$
for each $T_i \in T$ **do**
  if T_i for MacroVehicle **then**
    $m \leftarrow M_{macro}$
    $c \leftarrow C_{macro}$
    $U^m \leftarrow \text{MacroNav}(m, c)$
  else if T_i for MicroVehicle **then**
    $k \leftarrow \text{GetSubMapID}(T_i)$
    $m \leftarrow M^k_{micro}$
    $c \leftarrow C_{micro}$
    $U^c \leftarrow \text{MicroNav}(m, c)$
  else
    Other Command
  end if
end for

Concurrently, the MicroVehicle generates multiple detailed maps $\{M^j_{\text{micro}} \mid j = 1, \ldots, n\}$ in different small-scale environments using MicroSLAM on LiDAR/RGBD inputs C_{micro}

$$M_{micro} \leftarrow \{M^1_{micro}, \ldots, M^n_{micro}\} \leftarrow \text{MicroSLAM}(C_{micro}) \tag{4}$$

This hierarchical framework allows LLMs to abstract high-level intent, while the navigation controller grounds it into executable control actions. The pseudocode for this procedure is provided in Algorithm 1.

3.3. Hardware Platform

We develop a MacroVehicle and MicroVehicle platform. The MacroVehicle is designed for high-speed movement and heavy payload capacity, while the MicroVehicle is a lightweight, low-cost mobile platform, as illustrated in

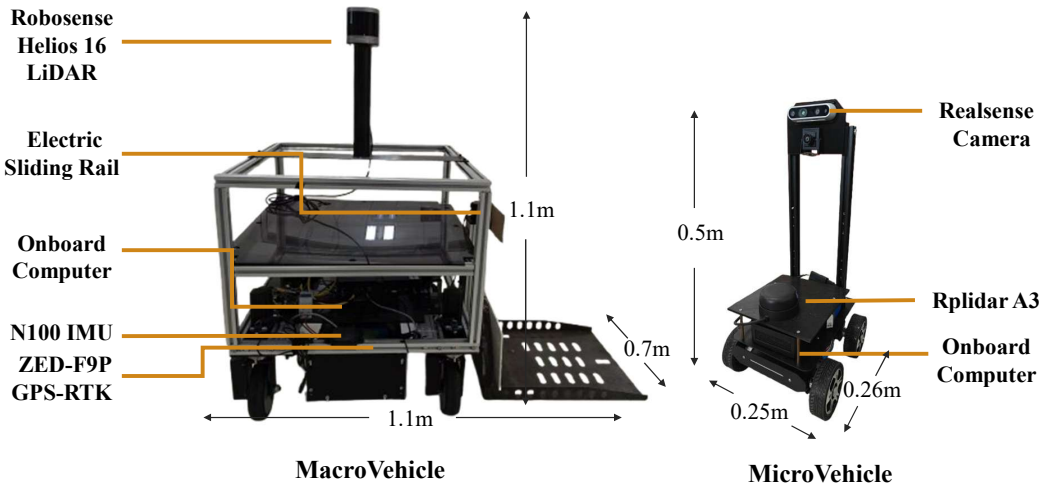


Figure 3: **MMVehicles platform.** The platform consists of a MacroVehicle and a MicroVehicle system. The MacroVehicle is equipped with a Robosense Helios 16 LiDAR, N100 IMU, and ZED-F9P GPS-RTK module for environmental perception and state estimation. An electric sliding rail with a loading platform facilitates the transportation and docking of the MicroVehicle. The MicroVehicle incorporates a RealSense D435i camera and Rplidar A3 sensor for small-scale environment sensing and state monitoring.

Fig. 3. The MacroVehicle serves as the central node, managing MicroVehicles with one-to-one and one-to-many cooperative control.

The MacroVehicle integrates an omnidirectional four-wheel drive system with a lightweight aluminum alloy frame, allowing easy modification and space adjustment. Each wheel is powered by two integrated drive units (IDUs) with brushless DC (BLDC) motors, achieving a top speed of 15 km/h. It also includes an aluminum docking platform equipped with an electric sliding rail for smooth MicroVehicle boarding and disembarking. The sensor suite consists of a Robosense Helios 16 LiDAR (360°×30° field-of-view, 10 Hz scanning rate), an N100 IMU (100 Hz attitude updates), and a ZED-F9P GPS-RTK module for perception and navigation. The main control system, based on an industrial computer (Intel i7 CPU + Nvidia GeForce RTX 3070 GPU), processes sensor data and runs the *Kangaroo* algorithm, while an STM32 microcontroller ensures precise omnidirectional motion control.

The MicroVehicle is a four-wheel differential drive robot weighing only 3.3 kg, achieving a top speed of 1.2 m/s. Its main control system, based on a Raspberry Pi 4B, processes sensor data and executes algorithms, while motion control is handled by the openCTR H60 controller. The MicroVehicle

is equipped with an Intel RealSense D435i depth camera and a RPLidar 2D LiDAR for environmental perception and navigation.

By integrating Virtual Private Network (VPN) technology with a remote Robot Operating System (ROS) bridge, the system enables stable communication and seamless data synchronization across different networks between the MMVehicles, ensuring reliable command execution under complex conditions.

3.4. IF-Planner

In complex and open environments, traditional control strategies based on predefined rules are inadequate to handle the infinite expansion of the task space and the uncertainties of dynamic environments. HMAS needs to dynamically plan its motion trajectories based on task descriptions and achieve joint optimization through collaborative coordination among multiple robots. To this end, we propose an iterative Feedback LLMs instruction primitive scheduling approach, designed to enhance the reliability and adaptability of task planning while enabling multi-task navigation for the Macro-Micro Vehicles across diverse physical environments. This approach consists of three key components: Interaction Module, Rule Validator, and LLMs Primitive Generator, as illustrated in Fig. 4.

Interaction Module employs an iterative feedback (IterFB) mechanism alongside a prior knowledge base to resolve ambiguities in natural language instructions and accurately interpret human intent. It dynamically clarifies task requirements and validates instruction rationality by integrating semantic information with environmental modeling. When instructions are ambiguous, mismatched, or unclear, the module requests clarification from humans through dialogue. This iterative process ensures precise task understanding, thereby providing high-quality input for task decomposition and instruction generation.

LLMs Primitive Generator integrates task descriptions with a structured prior knowledge database to generate motion primitive sequences using large language models, enabling the coordinated control of MMVehicles. To address the diverse task requirements across physical environments, this module decouples tasks into three hierarchical components: Task Agents, Skills, and Task Data. Task Agents represent the execution units within multi-agent systems, Skills encapsulate the executable robot operations, and Task Data provides an abstract representation of the environmental state. This component semantically parses task instructions and dynamically allocates

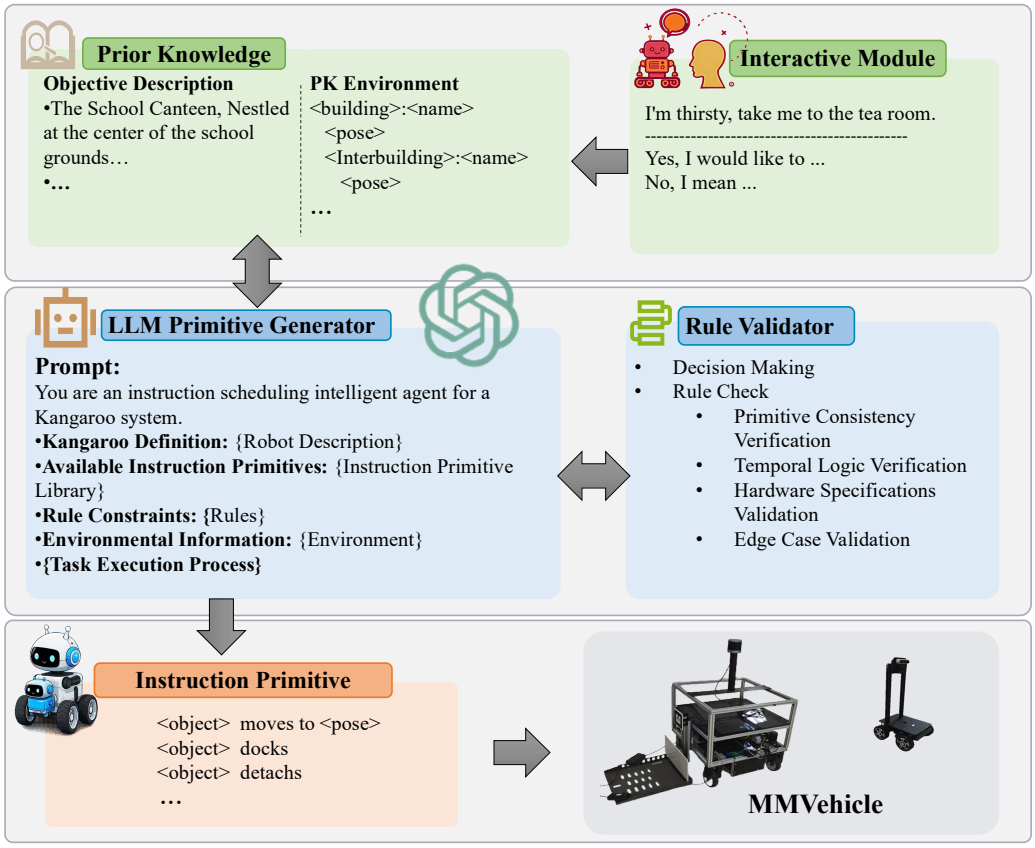


Figure 4: **Architecture of the IF-Planner.** The architecture comprises three components: (1) **Interactive Module**, which accurately interprets human intent and converts instructions into structured prompts; (2) **LLMs Primitive Generator**, which leverages LLMs to generate high-level instruction primitives based on inputs and prior knowledge; and (3) **Rule Validator**, which guarantees task feasibility and compliance with system constraints via rule-based reasoning. An iterative feedback mechanism ensures the correctness of generated motion primitives.

temporally sequenced task units by incorporating the heterogeneous characteristics of agents, such as variations in capabilities, sensor configurations, and motion constraints. This ensures the generation of executable action sequences that fulfill both the task objectives and multi-agent coordination requirements.

Rule Validator serves as the core validation module of the system, ensuring the safety, continuity, and precision of the generated instruction sequences. We propose a hierarchical validation framework comprising four

key components: (1) Primitive format consistency verification, (2) Kinematic feasibility check based on hardware specifications, (3) Temporal logic-based consistency verification for sequential command validation, and (4) Edge-case evaluation for boundary condition handling. The system features iterative feedback (IterFB) that detects and corrects anomalous command primitives through continuous feedback loops, thereby maintaining operational integrity while enforcing strict safety constraints across all execution phases.

3.5. MMVehicles Navigation Controller

The MMVehicles Navigation Controller introduces a cross-environment multi-scale navigation framework driven by instruction primitives generated for HMAS. The framework is divided into two main modules: (1) MacroNav, optimized for large-scale open environments, employs 3D LiDAR-based multi-sensor fusion to achieve real-time SLAM. MacroNav is designed to handle fast-moving tasks and large-area mapping, making it ideal for high-speed traversal by MacroVehicles. (2) MicroNav, tailored for small-scale, constrained environments, uses vision-LiDAR fusion and LLM-driven multi-map management. This combination allows MicroNav to perform cost-effective navigation in cluttered spaces, focusing on lightweight SLAM and semantic-aware environment adaptation, ideal for micro vehicles with high maneuverability.

Both navigation systems consist of two components: the SLAM module and the navigation module. The SLAM module enables real-time self-localization and environment mapping, formally defined as a recursive optimization problem for state estimation and map building. It addresses challenges such as solving $p(x_t|x_{t-1}, \mu_t, z_t, m)$ and $p(m|x_t, z_t)$. The autonomous navigation module uses the robot's map m, current position data x_t, control commands u_t, and sensor data z_t to plan the optimal path:

$$P^* = \arg\min_P \int_{t_0}^{t_f} \mathcal{C}(x_t, u_t, m, z_t, t)dt, \tag{5}$$

while satisfying kinematic and obstacle avoidance constraints to reach the target position. Subsequent sections will detail the technical implementation and performance optimization of this system.

MacroNav: This study presents a 3D LiDAR-based SLAM framework for efficient localization and navigation in large-scale open environments. Building on Fastlio2[44], the system integrates GNSS-assisted global point

cloud matching re-localization with ICP-based incremental pose tracking. A hierarchical localization approach ensures global map consistency through factor graph optimization and enhances local motion estimation with IMU data. The system generates dual-modal maps, including high-fidelity 3D point clouds and lightweight 2D grids. To reduce computational load, 3D point clouds are converted into 2D occupancy grids via ground segmentation and projection transformations. Integrated with the ROS navigation framework, the system enables cost map generation, global path planning, and local trajectory optimization, improving real-time navigation performance in complex environments.

MicroNav: The high cost of 3D LiDAR and the lack of GNSS availability in indoor environments make it impractical to deploy micro vehicles using the same approach as MacroVehicles. To address this, we enhance Cartographer[21] by integrating 2D LiDAR, IMU, and RGB-D camera, leveraging projection transformation to fuse 3D point clouds with 2D LiDAR scans. This improves 3D spatial perception and enables robust navigation in complex environments. In practice, micro vehicles often operate in diverse environments, and different types may be deployed for specialized tasks. To manage this, MicroNav introduces an efficient multi-map system that represents the environment as a set of 2D grid maps, $M = \{m_1, m_2, \ldots, m_k\}$, using a discretized storage strategy. An LLM-driven framework ensures real-time adaptive map selection for seamless navigation. For localization and path planning, we use an AMCL algorithm that fuses visual and LiDAR data, achieving centimeter-level tracking and obstacle avoidance. The path planning approach in MicroNav is consistent with that of MacroNav.

MMVehicles Coordination: The macro–micro vehicle coordination module is designed with a four-level architecture.

- **Task planning level:** the IF-planner dynamically decomposes tasks by monitoring the cooperative states of MacroVehicles and microVehicles in real-time, and performs adaptive task allocation through multi-dimensional capability evaluation, ensuring that both types of intelligent agents efficiently execute cross-environment tasks considering their working limits and special skills.

- **Communication level:** we employ a dual-channel hybrid networking architecture with context-aware protocol switching. In the docked configuration, the system employs TCP-based centralized task scheduling utilizing a standardized command schema comprising task type

identifiers, execution parameter vectors, and status feedback matrices. During detected operations, a "ROS + VPN" framework establishes authenticated communication channels, enabling secure real-time data interchange between macro-micro vehicular agents distributed across heterogeneous network domains while maintaining strict Quality of Service guarantees for global cooperative mission execution.

- **Collaborative perception level:** we employ a multi-source map reference point alignment method to achieve coordinated map selection and pose initialization between MacroNav and MicroNav systems, ensuring spatial consistency between Simultaneous Localization and Mapping (SLAM) systems and semantic navigation waypoints.

- **Physical docking level:** the docking layer serves as the core subsystem for MMVehicles physical coupling. The system implements a dual-stage positioning strategy based on heterogeneous sensor data fusion, combining visual feature recognition for coarse positioning with LiDAR point cloud plane tracking for fine alignment. This approach enables real-time relative pose estimation and employs PID control algorithms to establish a closed-loop actuation mechanism, ultimately achieving millimeter-precision docking operations.

This architecture effectively addresses challenges in task allocation, environmental perception, and precise execution under constrained communication conditions.

4. Experiment

We evaluate *Kangaroo* in real-world environments using our self-developed MMVehicles platform. In Section 4.1, we describe the experimental setup and evaluation metrics. Section 4.2 assesses the SLAM-based navigation performance of both the MacroVehicle and MicroVehicle. In Section 4.3, we evaluate the impact of different large language models and the rule-based Validator on motion primitive generation within the IF-Planner module. Finally, Section 4.4 provides a comprehensive evaluation of the system under different levels of demand across various cross-environment scenarios, demonstrating the effectiveness of the proposed approach.

4.1. Experiment Setup

4.1.1. ***Experimental Scenarios***

We conduct real-world experiments using our self-developed *MacroVehicle* (1.1m width) and *MicroVehicle* (0.25m width) in two representative cross-environment navigation scenarios, quantitatively evaluating the system's adaptability and robustness under heterogeneous spatial conditions:

- **Large-scale indoor to small-scale indoor (VastHall–TinyRoom):** The robot transitions from a structured, open indoor environment exceeding $500m^2$ with an obstacle density below $0.1/m^2$, into a confined, semantically rich space of less than $20m^2$ with an obstacle density up to $2/m^2$. The navigable path narrows abruptly from $3m$ to $0.5m$, rendering it impassable for the MacroVehicle.

- **Outdoor to indoor (OutdoorRoad–NarrowPath):** The robot moves from an unstructured outdoor environment with GPS availability, a path width over $4m$, and irregular obstacles such as trees and parked vehicles, into a narrow $0.5m - wide$ indoor corridor where GPS is unavailable.

4.1.2. ***Dataset***

To support research on demand-driven motion primitive generation, we construct the IF-Planner dataset. Specifically designed for cross-environment navigation scenarios, the dataset covers 9 main buildings (each containing 5-7 internal buildings), with MacroVehicle navigation between buildings and microVehicle navigation within internal buildings, totaling 64 typical indoor and outdoor scenarios. Each data sample includes the agent's current position, natural language task description, and corresponding ground-truth motion goal, providing a systematic benchmark for evaluating navigation algorithms under varying task complexities and heterogeneous platform configurations.

We define four task categories, each with 100 test samples, resulting in 400 annotated test samples:

- **DS:** A direct demand with a single target. Example: *"Can you help me go to the canteen?"*

- **DM:** A direct demand with sequential goals. Example: *"I need to play basketball at the Sports Center, then do laundry in the Dormitory."*

Symbol	Description
SR	The success rate of tasks completed.
NE	Navigation error between the final and target positions.
OE	Orientation error between the final and target orientations.
MPE	Measured position error between estimated and actual positions.
σ_{NE}	Standard deviation of navigation error.
σ_{OE}	Standard deviation of orientation error.
σ_{MP}	Standard deviation of measured position error.
N_{attempt}	Average number of attempts required for task success.
T_{nav}	Average navigation time per meter.
D-N	Whether demand-condition navigation is supported.
C-N	Whether cross-environment navigation is supported.

Table 2: Metrics and their descriptions

- **IS:** An indirect demand implying one target. Example: *"I need to borrow a copy of 'The Great Gatsby'."*
- **IM:** An indirect demand requiring reasoning to infer multiple goals. Example: *"I want to cook by myself today. I need to buy some vegetables and meat, then go back to the dormitory to try cooking."*

4.1.3. ***Evaluation Metrics***

We adopt the following metrics to assess navigation performance as shown in Table 4.1.3.

4.2. MMVehicles Performance

To evaluate the navigation performance of the MacroVehicle and MicroVehicle in their respective adaptive scenarios, we conduct experiments using our self-developed Macro-Micro Vehicles system in a real-world cross-environment setting, VastHall-TinyRoom. Each experiment consists of four stages: (1) SLAM-based mapping, (2) prior knowledge integration, (3) IF-Planner instruction generation, and (4) navigation execution. Since the primary objective of this section is to evaluate the individual navigation performance of each vehicle, we eliminate the influence of decision-making algorithms by directly employing the correct motion primitives generated by IF-Planner. As shown in Fig. 5, the MacroVehicle operates in the VastHall

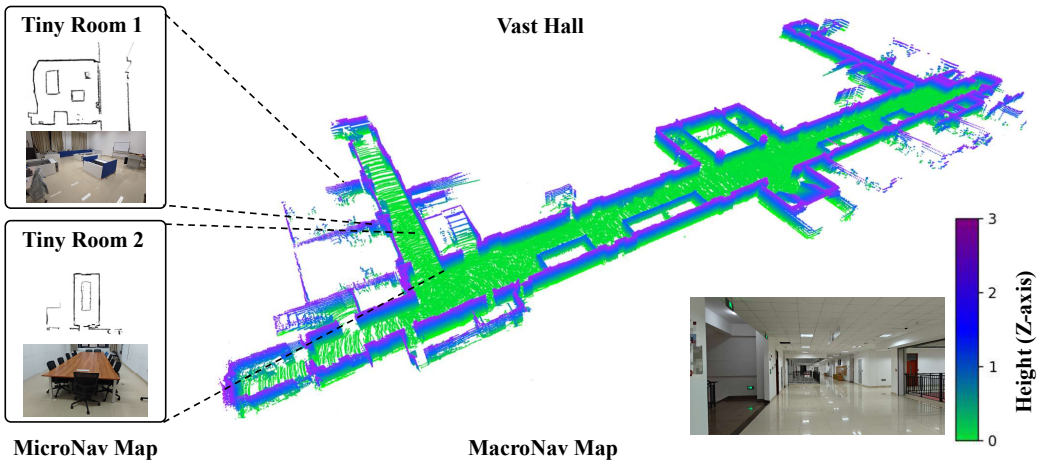


Figure 5: **Kangaroo Navigation Maps.** The left panel shows the 2D grid maps generated by the MicroVehicle in two "TinyRoom" scenarios using 2D SLAM algorithms. The right panel illustrates the 3D point cloud map constructed by the MacroVehicle in the "VastHall" environment using 3D SLAM algorithms, where color depth represents building height variations.

using MacroNav to build a 3D point cloud map, while the MicroVehicle navigates the TinyRoom with MicroNav to generate a 2D grid map. To evaluate performance, we define ten goals in each scene and conduct fifteen navigation rounds, yielding 150 samples for analysis.

As shown in Table 3, the results demonstrate that *Kangaroo*, integrating both MacroNav and MicroNav, achieves consistently high success rates across multi-scale environments. MacroNav exhibits robust navigation capabilities with a success rate of 92.7%. It effectively handles challenges such as sparse object distribution and long traversal distances while satisfying the success criteria. Its average position error (0.17 m) and orientation error (0.95°) indicate its ability to maintain reasonable accuracy in expansive spaces. In contrast, MicroNav achieves a perfect success rate of 100% with exceptionally low position (0.04 m) and orientation (0.04°) errors, demonstrating its precision and reliability in structured, confined spaces. The complementary strengths of MacroNav and MicroNav highlight the versatility of *Kangaroo*, ensuring high performance and adaptability across different scales, making it a robust solution for multi-scale navigation tasks.

Method	SR ↑	NE ↓	$\sigma_{\mathbf{NE}}$ ↓	OE ↓	$\sigma_{\mathbf{OE}}$ ↓	MPE ↓	$\sigma_{\mathbf{MP}}$ ↓
MacroNav	92.7%	0.17	0.07	0.12	0.07	0.03	0.26
MicroNav	100%	0.04	0.01	0.04	0.03	0.02	0.02

Table 3: Experimental Performance of MacroNav and MicroNav.

4.3. Performance of different LLMs in IF-Planner

To evaluate the effectiveness of IF-Planner and its generalizability across different LLMs, we conduct both ablation and comparative experiments on four types of tasks from the IF-Planner dataset (as introduced in Section 4.1), using **GPT-4o**[45], **Deepseek-Chat**[46], **QWQ-32B**[47], and **Deepseek-Reasoner**[48].

In the motion primitive generation experiments with LLMs, we set the maximum number of iterative feedback rounds to ten. We compare the performance of four LLMs on four task types by analyzing how the success rate evolves with the number of attempts. As shown in Fig. 6, the success rates of all models increase progressively with more attempts, validating the effectiveness of the iterative framework. Tasks described directly typically achieve near-perfect success rates within five attempts, while those expressed indirectly stabilize after five iterations. In terms of model performance, Deepseek-Reasoner demonstrates the best results, outperforming QWQ-32B, followed by Deepseek-Chat and GPT-4o. Even in the most challenging tasks, the weakest model achieves a success rate exceeding 75% after five attempts, further confirming the robustness and generalizability of the method.

Table 4.3 presents a detailed comparison of the four LLMs, reporting success rates and the average number of attempts for successful cases, both with and without iterative feedback. Deepseek-Reasoner consistently outperforms the other models, achieving over 96% success across all tasks, indicating the high accuracy of IF-Planner in motion primitive generation. The benefit of iterative feedback is particularly evident in indirect tasks, where it leads to performance gains of up to 60%. Additionally, the average number of attempts remains below three in all settings, demonstrating the efficiency of IF-Planner in responding to demand-driven instructions.

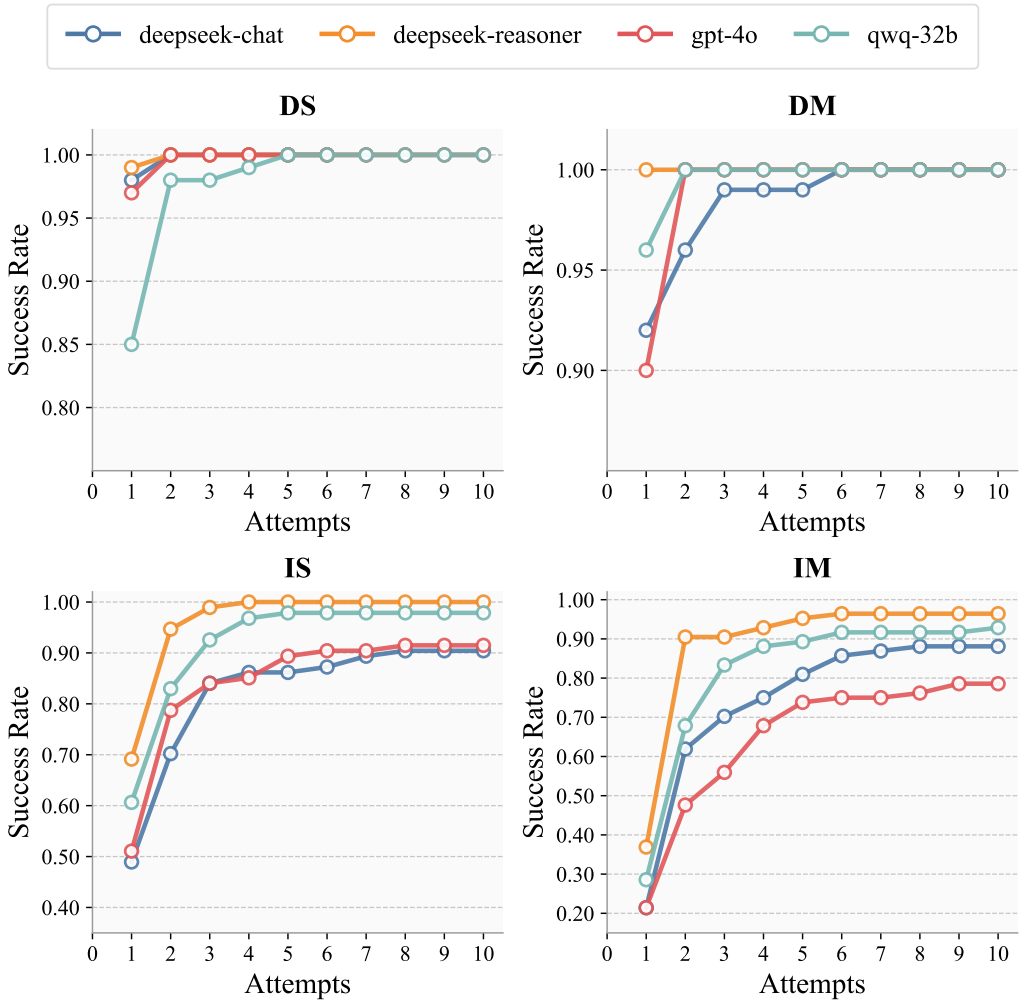


Figure 6: **Success Rate Trends of Different LLMs in Four Task Types Across Attempts.** All models show clear performance improvements as the number of attempts increases, demonstrating the effectiveness of our iterative feedback mechanism in guiding motion primitive generation.

4.4. Overall Navigation Performance in Different Cross-Environments

To evaluate the overall performance of *Kangaroo*, we conducted real-world experiments across two cross-environment settings and four task types (Section 4.1). These experiments compare *Kangaroo* against state-of-the-art methods under diverse scenarios and task demands, assessing its effectiveness

Task Type	Model	SR↑ (w/o FB)	SR↑ (w/ FB)	$N_{\text{attempt}} \downarrow$
DS	GPT-4O	97%	100%	1.03
	QWQ-32B	85%	100%	1.20
	Deepseek-Chat	98%	100%	1.02
	Deepseek-Reasoner	**99%**	**100%**	**1.01**
DM	GPT-4O	90%	100%	1.10
	QWQ-32B	96%	100%	1.04
	Deepseek-Chat	92%	100%	1.15
	Deepseek-Reasoner	**100%**	**100%**	**1.00**
IS	GPT-4O	51%	91%	1.78
	QWQ-32B	61%	98%	1.60
	Deepseek-Chat	49%	90%	1.89
	Deepseek-Reasoner	**69%**	**100%**	**1.37**
IM	GPT-4O	21%	79%	2.73
	QWQ-32B	29%	93%	2.21
	Deepseek-Chat	21%	88%	2.53
	Deepseek-Reasoner	**37%**	**96%**	**1.79**

Table 4: Performance Comparison of LLMs in Primitive Generation Across Four Task Types.

in hierarchical multi-agent navigation.

We designed two representative environments: (1) *VastHall-OutdoorRoad* for large-scale open navigation, and (2) *TinyRoom-NarrowPath* for constrained indoor-like tasks. Fig. 7 illustrates both environments and complete navigation process of MMVehicles, including IF-Planner-generated motion primitives and coordinated Macro-Micro Vehicles movements. Navigation targets were abstracted as buildings (for open environments) and internal buildings (for constrained spaces). Considering battery consumption and motor wear, we fixed navigation speeds at $\frac{1}{3}$ of maximum speed (1.4 m/s for the MacroVehicle and 0.4 m/s for the MicroVehicle).

Our evaluation compares *Kangaroo* with four baseline methods: **CartoNav**, which uses only the MicroVehicle for navigation, integrated with Cartographer and ROS; **DDN** and **FastNav**, which operate in their respective

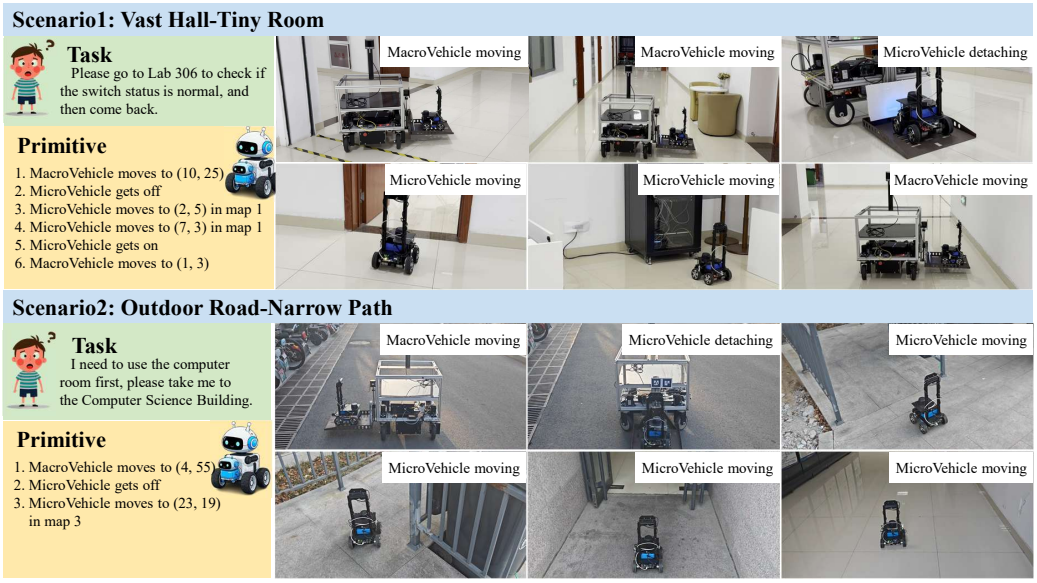


Figure 7: **Experiment Scene for Kangaroo.** The figure illustrates the experimental setup and navigation flow for *Kangaroo*, presenting two distinct scenarios: Vast Hall-Tiny Room and Outdoor Road-Narrow Path. It includes the movement process of the MacroVehicle, as well as the detaching, moving, and docking operations of the MicroVehicle.

environments, similar to our DirectSingle-Goal tasks and even IndirectSingle-Goal tasks; and **LLM-Nav**, which adopts the same architecture as *Kangaroo* but directly generates motion primitives through Deepseek-Reasoner without incorporating a feedback mechanism.

As shown in Table 5, *Kangaroo* demonstrates robust and consistent performance across diverse scenarios. In VastHall-TinyRoom, it achieves success rates above 85%, and even in the more challenging OutdoorRoad-NarrowPath, its success rate remains above 78%, showcasing strong adaptability to complex environments. In contrast, CartoNav[21] performs well in structured environments like VastHall-TinyRoom, achieving 100% success in DirectSingle-Goal tasks. However, it lacks cross-environment adaptability and demand-driven navigation capabilities. This inefficiency becomes even more pronounced in large-scale or long-distance deployments, where the lack of flexibility and adaptability severely limits its practical applications. "Find What You Want"[1] and FastNav[3] support demand-driven navigation but struggle with cross-environment adaptability. The former achieves only 19.3% success in IndirectSingle-Goal tasks, showing significant performance instability and

Scenario	Method	D-N	C-N	T_{nav} (s/m)	SR↑ DS	DM	IS	IM
VastHall & TinyRoom	CartoNav[21]	✗	✗	2.73	100%	-	-	-
	DDN[1]	✓	✗	-	21.5%	-	19.3	-
	FastNav[3]	✓	✗	-	70%	-	-	-
	LLM-Nav (ours w/o FB)	✓	✓	1.39	87%	88%	61%	33%
	Kangaroo (ours)	✓	✓	**1.39**	**88%**	**88%**	**88%**	**85%**
OutdoorRoad & NarrowPath	CartoNav[21]	✗	✗	2.73	60%	-	-	-
	LLM-Nav (ours w/o FB)	✓	✓	1.39	80%	81%	56%	30%
	Kangaroo (ours)	✓	✓	**1.39**	**81%**	**81%**	**81%**	**78%**

Table 5: Comparison of Kangaroo with Advanced Methods Across Two Scenarios and Four Tasks.

struggling to handle more complex navigation scenarios. LLM-Nav performs adequately in direct tasks, but experiences a significant drop (over 50%) in performance for complex demand-driven indirect tasks. This highlights the critical role of our validation feedback mechanism in ensuring reliable navigation across diverse scenarios.

These comprehensive experimental results validate the superior efficiency and adaptability of *Kangaroo*. By supporting both demand-driven navigation and cross-environment adaptability, *Kangaroo* consistently outperforms existing methods. Furthermore, our heterogeneous MMVehicles architecture is essential for practical applications that require flexibility, scalability, and robustness.

4.5. Discussion

We comprehensively evaluate the performance of IF-Planner, the MMVehicles Navigation submodule, and the overall capabilities of *Kangaroo*, covering two cross-environment scenarios and four demand-level tasks. Through meticulously designed experiments and in-depth analysis, we effectively demonstrate the accuracy in interpreting demand instructions and the robustness and efficiency in executing complex cross-environment navigation tasks.

The current system still exhibits limitations. First, the types of mobile robots involved in the system are limited. For example, due to their in-

sufficient terrain adaptability, they may be unable to operate effectively on extremely rugged terrains. Second, the Macro-Micro Vehicles have limited differences in mobility and flexibility. Although the MacroVehicle achieves higher speed than the MicroVehicle, its efficiency in ultra-long-distance tasks remains insufficient. Future research will expand to include more diverse forms such as quadruped robots, unmanned vehicles, and drones, along with varied sensor configurations, to enhance the system's adaptability in complex scenarios like highways, tunnels, and disaster sites. Additionally, we plan to explore more heterogeneous combinations, such as a wheeled carrier vehicle equipped with a quadruped robot or a tracked carrier vehicle carrying a wheeled robot to achieve functional complementarity and collaborative optimization.

At the system architecture level, although the *Kangaroo* platform theoretically supports complex topologies such as "1-N" and "N-N", existing experiments focus primarily on "1-1" scenarios. As the number of agents increases, resource allocation within edge computing networks and dynamic task scheduling will face combinatorial explosion challenges, necessitating further optimization of distributed coordination algorithms and adaptive resource management strategies.

5. Conclusion

We propose a novel navigation system named *Kangaroo*, a heterogeneous multi-agent navigation approach guided by *IF-Planner*, designed to enable demand-driven cross-environment navigation. *Kangaroo* is the first to address the challenge of executing diverse natural language-based navigation demands in environments characterized by multi-scale continuity. Our method comprises two core modules: (1) the IF-Planner module, which handles task understanding, rule-based validation, and motion primitive generation and scheduling. It integrates an external knowledge base to enhance commonsense reasoning and environmental abstraction; and (2) the MMVehicles navigation control module, which handles mapping, navigation, and coordination between the Macro-MicroVehicles to ensure accurate arrival at the goal. Experimental results demonstrate that *Kangaroo* consistently achieves high success rates across various multi-scale environments. By integrating LLMs with HMAS, we effectively bridge the gap between high-level task planning and low-level execution, enabling robust navigation across environments of varying spatial scales.

Our system establishes a research foundation for human-centric collaborative operations among HMAS and is expected to advance the practical application of autonomous operation technologies for unmanned systems in open environments.

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